Visual Studio LIVE! | Las Vegas
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

SQL Server Indexing

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Level: Intermediate

About Me

- Denny Cherry & Associates Consulting
- 6 books
- Dozens of articles
- Microsoft MVP
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Microsoft Most Valuable Professional | Microsoft CERTIFIED Master SQL Server 2008 | vmware vEXPERT

Today’s Goals

- Introduce the different kinds of indexes
- Common Misconceptions about indexes
- Downsides to indexes
- Introduce advanced index tuning techniques
- Q & A

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Different Kinds of Indexes

Several Kinds of Indexes

- Clustered
- Non-clustered
- Full Text
- XML
- ColumnStore Indexes

Clustered Indexes

1 Clustered Index per table

Is the base table

Up to 16 indexed columns can be part of the index

- (15 if the table contains any XML indexes)

Primary Key will by default be the Clustered Index

Must be created on the same filegroup as the table

Clustered Indexes should be as narrow as possible

While not required, they are highly recommended

Non-clustered Index

Up to 999 per table Starting with SQL Server 2008

- 255 in SQL Server 2005 and below

Up to 16 indexed columns in the index

Non-indexed columns can be included via INCLUDE statement

Non-Clustered indexes always contain the clustered index columns (when table has a clustered index)

When table is a heap, the Row ID is stored in every non-clustered index.

Can be created on any filegroup within the database

Can be filtered indexes to include fewer rows in the index.

Differences between unique and non-unique clustered indexes

Non-Unique clustered indexes have an extra column called the uniqueifier which ensures that values within the index are unique.

Uniqueifier is only used for rows which are not unique.

EmpId	Uniqufier
1	
2	
3	
4	0
4	1
5	
6	
7	0
7	1
8	

Full Text Indexes

Not accessed via normal SELECT statements

Require use of a predicate:

- CONTAINS
- CONTAINSTABLE
- FREETEXT
- FREETEXTTABLE

Can be used to search binary values (doc, docx, xls, pdf) stored within the database.

Natural Language Search

Can index XML documents, but only indexes the values, not the tags.

Full Text Indexes (SQL 2005 and below)

Created and managed outside of the database via Microsoft Search Service

Backed up with the database (starting in SQL 2005)

Searches entire index and returns all matches, which you then filter against your normal table to return correct set of rows.

Full Text Indexes (SQL 2008 and up)

Now stored within the database

Command is still parsed via MS Search service, but looking is done natively

Full text search now only searches the required subset of rows

When creating your indexes use an identity field as the key to improve query performance.

XML Indexes

- Allows you to index specific nodes of the XML document
- 249 XML Indexes pre table
- Requires a Clustered Index on the table
- Each xml column can have a single primary XML index and multiple secondary XML indexes
- XML Indexes can only be created on a single XML Column
- No online rebuilds
- Not available for XML variables. Only used on tables.

Primary XML Index

When created creates a hidden node table

- Contains base table primary key and 12 columns of info about every node within the XML value

Effectively the clustered index on the node table

- Base Table Clustered Index Value
- Node id from the node table

Increases storage

- 200-500%

Secondary XML Indexes

Non-Clustered Indexes on the hidden node table

Three kinds of secondary indexes

- PATH index on the node id (path) and the value
- VALUE index is on the value and the node id (path)
- PROPERTY index is on the base table's clustered index, node id (path) and the value

What is a ColumnStore Index?

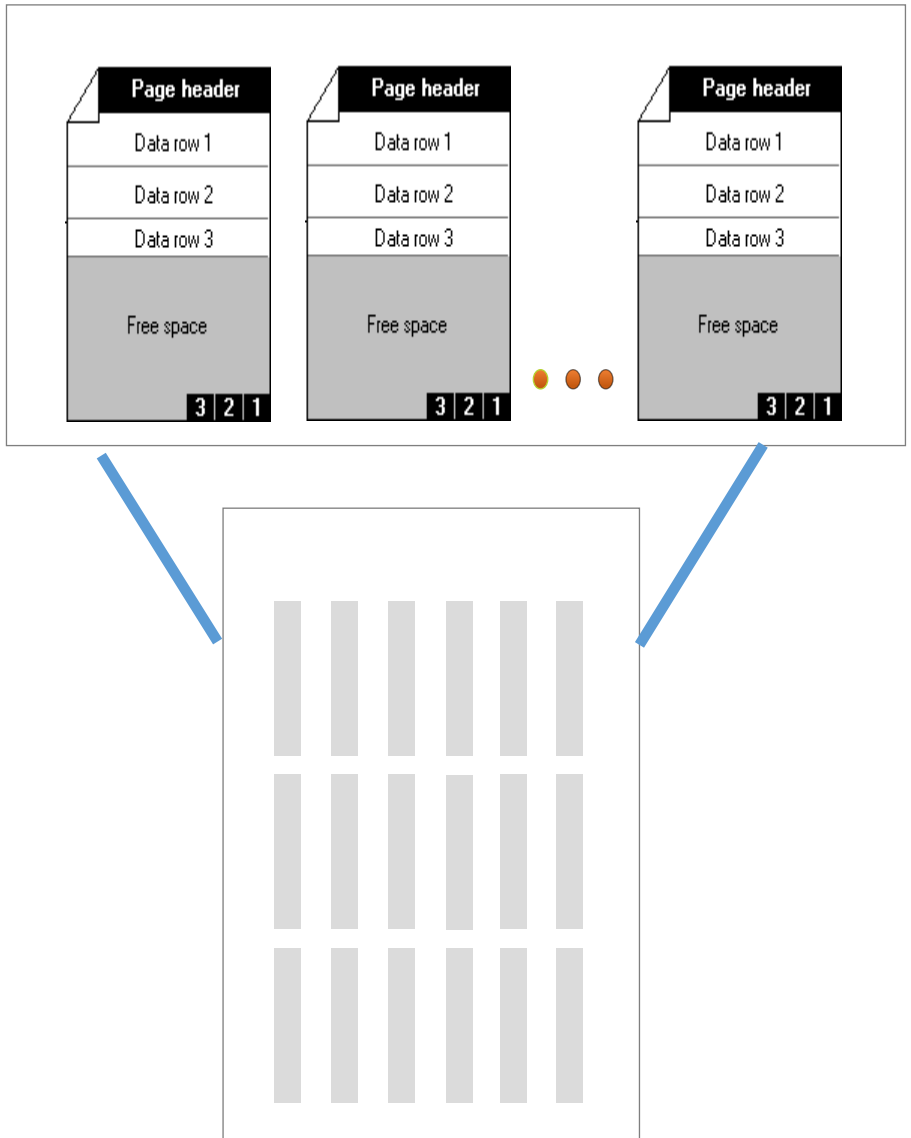
Totally new and different approach to indexing

Data is stored via columns not rows

Each column is stored separately, then compressed using VertiPak compression engine

SQL Server’s first B-Tree less index

How does ColumnStore do that?



Existing Pages

Needed Columns

ColumnStore: Use Case

Data continues to grow, but performance requirements stay the same

Many data warehouses approach PB ranges

Data needs to be filtered, aggregated, and grouped despite the size of the dataset

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Common Misconceptions about indexes

Indexes don’t require maintenance

If I create one index for each column in my where clause I’ll be fine

The table is sorted based on the order of the Clustered Index

Clustered Indexes are required

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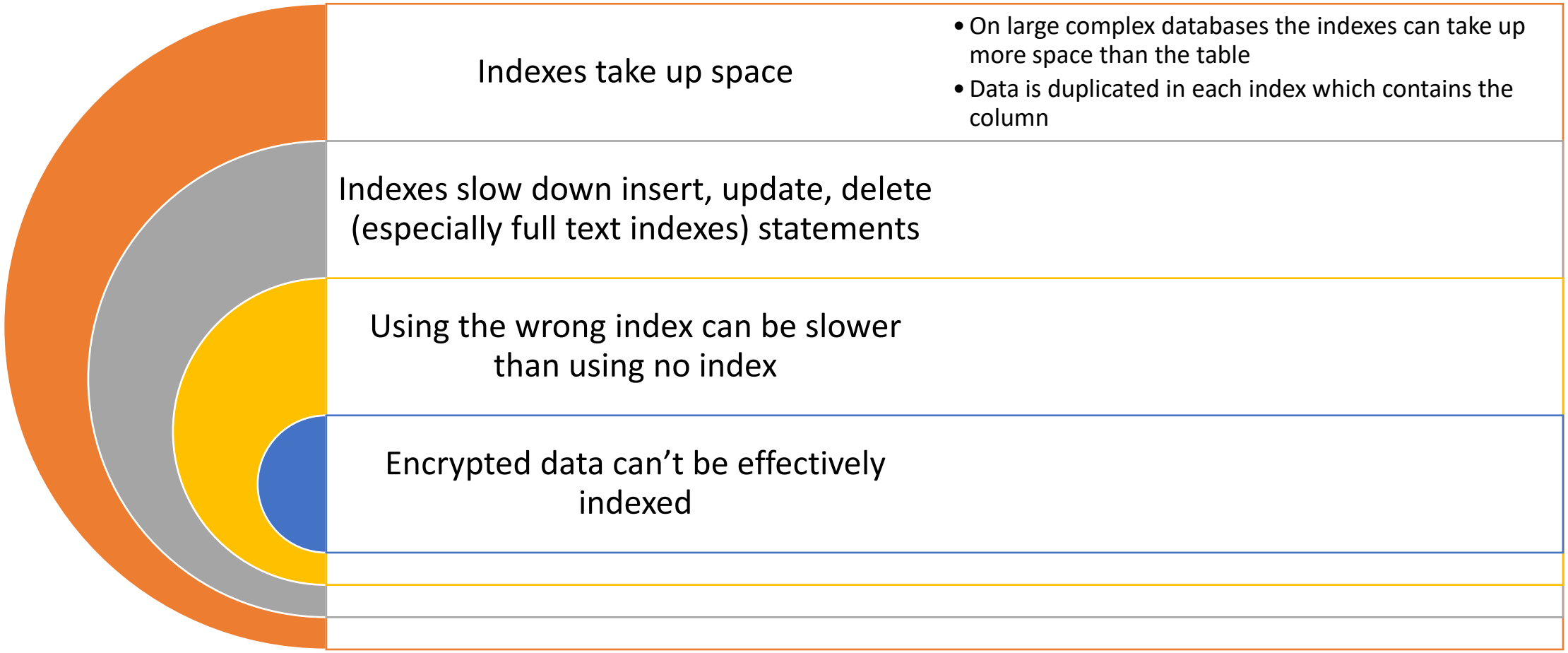
Downsides to indexes



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Advanced Index Tuning Techniques

Fillfactor

- Tells the SQL Server how much free space to leave in the leaf level pages.

Padding

- Tells the SQL Server to use the Fillfactor setting to leave free space in the intermediate-level pages.

Online Rebuilds

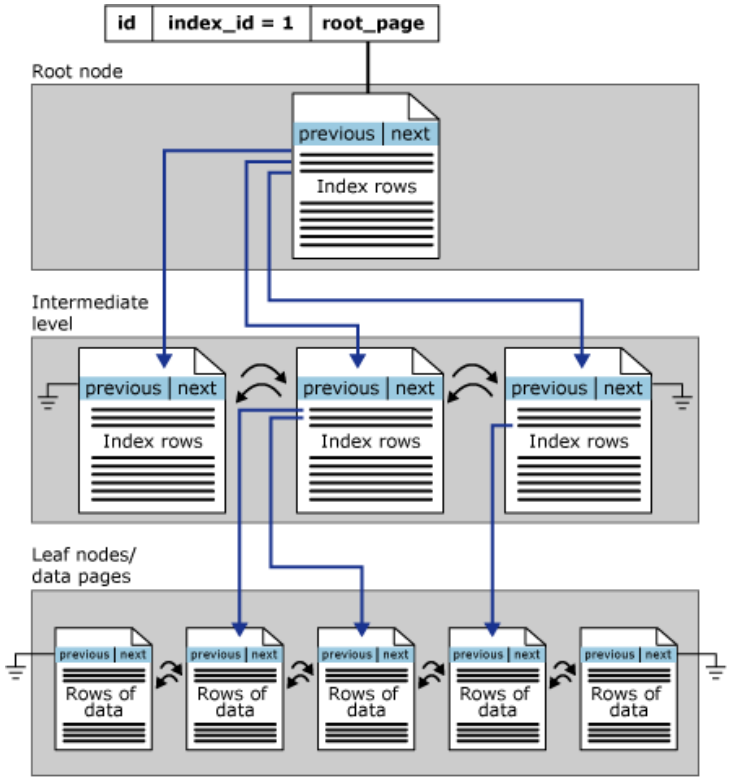
Data Compression

Using the Advanced Index Tuning Techniques

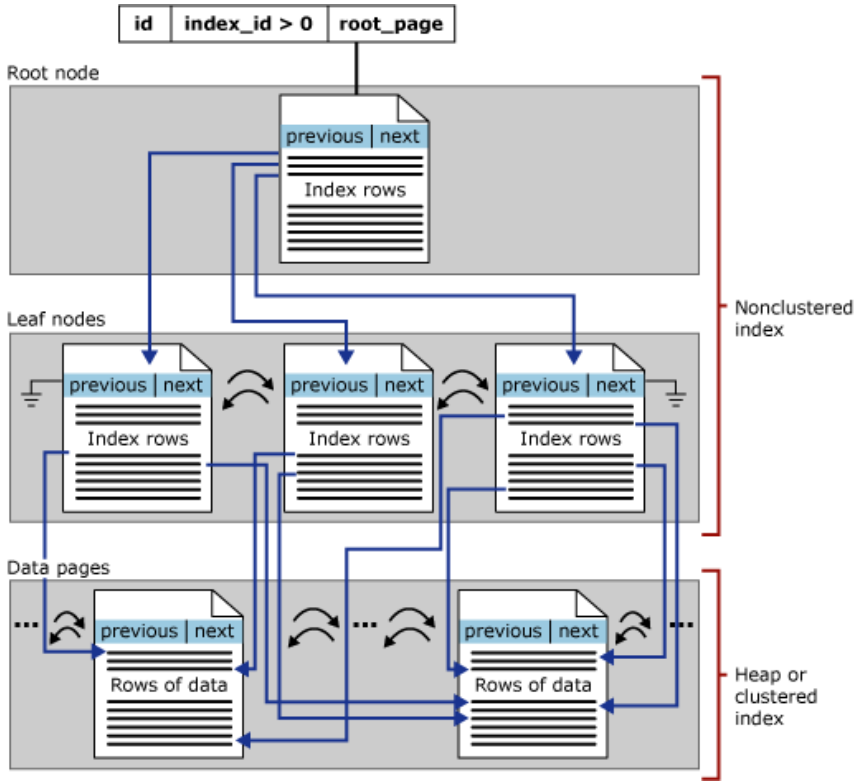
```
CREATE INDEX MyIndex ON dbo.MyTable
ON (Col1, Col5, Col3)
INCLUDE (Col4, Col2)
WHERE Col6 = ‘Value3’
WITH (FILLFACTOR=70, PAD_INDEX=ON, ONLINE=ON,
  DATA_COMPRESSION = ROW | PAGE);
```

Physical Index B-Tree Layout

Clustered (BOL 2005 / 2008)



Non-Clustered (BOL 2005 / 2008)



How large are my indexes?

- SELECT *
- FROM sys.dm_db_index_physical_stats (db_id(), object_id(‘table_name’), null, null, ‘detailed’)
  - Database Id
  - Object Id
  - Index Id
  - Partition Number
  - Mode (NULL | Limited, Sampled, Detailed)

What Indexes are being used?

```
DECLARE @dbid INT
  , @dbName VARCHAR(100);

SELECT @dbid = DB_ID()
  , @dbName = DB_NAME();

WITH partitionCTE (OBJECT_ID, index_id, row_count, partition_count)
AS
(
  SELECT [OBJECT_ID]
      , index_id
      , SUM([ROWS]) AS 'row_count'
      , COUNT(partition_id) AS 'partition_count'
  FROM sys.partitions
  GROUP BY [OBJECT_ID]
      , index_id
)

SELECT OBJECT_NAME(i.[OBJECT_ID]) AS objectName
      , i.name
      , CASE
        WHEN i.is_unique = 1
          THEN 'UNIQUE '
        ELSE ''
      END + i.type_desc AS 'indexType'
      , ddius.user_seeks
      , ddius.user_scans
```

Q & A

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